HUM 4747: Legal Issues and Cyber Law

E-Contracts and Digital Signatures: Legal & Security Perspectives

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Outline

What is a Contract?

Definition

A **contract** is a **legally binding agreement** between two or more parties that identifies the parties and describes the roles, responsibilities, time, and place of the agreement.

Key Roles

Offeror The person who **makes** an offer

Offeree The person who **receives** an offer

Legal Basis in BD

- **Contract Act 1872** — enacted by the Parliament of Bangladesh
- Governs general contract law

Why Contracts Matter for Software Engineers

- **Software licenses** are contracts (MIT, GPL, proprietary EULAs)
- **SaaS agreements** — Terms of Service you agree to
- **Freelance/consulting** — Scope, payment, deliverables
- **Employment agreements** — NDA, non-compete, IP ownership
- **Vendor contracts** — Cloud providers (AWS, Azure), APIs, SDKs
- **Open-source contributions** — Contributor License Agreements (CLAs)

Key Point

Every time you click "I Agree" on a Terms of Service — you are entering a contract.

What is an E-Contract?

Definition

Any kind of contract formed by the **interaction of two or more individuals using electronic means**, modelled, specified, executed and deployed by a software system.

Key Characteristics:

- Parties do **not meet physically** in most cases
- No physical boundaries — **cross-border** by nature
- No **handwritten signatures** — digital signatures are used
- Electronic records serve as **evidence in court**
- **Jurisdictional issues** are a major challenge in case of breach

Types of E-Contracts

Click-Wrap Agreements

- User clicks "**I Agree**" button
- Common in software installations, app downloads
- Example: Installing VS Code, signing up for GitHub

Browse-Wrap Agreements

- Terms accessible via **hyperlink** on the website
- Continued use = acceptance
- Legally **weaker** than click-wrap

Shrink-Wrap Agreements

- License terms inside **software packaging**
- Opening the package = acceptance
- Less common in the digital age

Electronic Data Interchange (EDI)

- Automated B2B transactions
- Structured electronic exchange
- Common in supply chain systems

E-Commerce and Software Licensing

License Type	Description	Example
Proprietary	Restrictive; source code not shared	Microsoft Office, Adobe CC
Open-Source (Permissive)	Free to use, modify, distribute	MIT, Apache 2.0, BSD
Open-Source (Copyleft)	Derivatives must remain open-source	GPL, AGPL
Freemium / SaaS	Free tier + paid features via subscription	GitHub, Slack, Figma

For Developers

Understanding license types is crucial when using third-party libraries. Using a **GPL library** in your proprietary product can force you to open-source your entire codebase!

International Legal Framework

UNCITRAL Model Law on E-Commerce (1996)

- Adopted by the **United Nations Commission on International Trade Law**
- Establishes rules that **validate and recognize** contracts formed through electronic means
- Provides a framework for countries to adopt e-commerce legislation

Bangladesh Status

Bangladesh has **not enacted** the UNCITRAL Model Law directly, but the **ICT Act 2006** incorporates many of its principles.

ICT Act 2006 — Key Provisions for E-Contracts

Section 6: Recognition of Electronic Records

Any record in electronic form is recognized as valid, provided the record is **accessible and usable for subsequent reference**.

Section 7: Legal Recognition of Digital Signatures

Digital signatures have **legal recognition** — electronic signing of records is valid.

Key Implications

- Handwritten signature is **not always needed** for a contract to be credible
- Contracts **cannot be refused** simply for being electronic
- **Section 11**: Government agencies have **no obligation** to accept electronic documents (public sector exception)

Documents That CANNOT Be Signed Electronically

Under current Bangladesh law, certain documents **require physical signatures**:

1. **Wills**
2. **Power of Attorney**
3. **Deed of Sale** for immovable property
4. Agreements where **stamp duty is payable**
5. Documents requiring signature before a **notary public** and/or witnesses

Cross-Border Consideration

When dealing with cross-border transactions, different countries have different rules. Always check the **Global eSignature Compliance Guide** for the relevant jurisdiction.

What is a Digital Signature?

Definition

A **digital signature** is a mathematical scheme for verifying the **authenticity** of digital messages or documents, using **public-key cryptography**.

A valid digital signature must be able to:

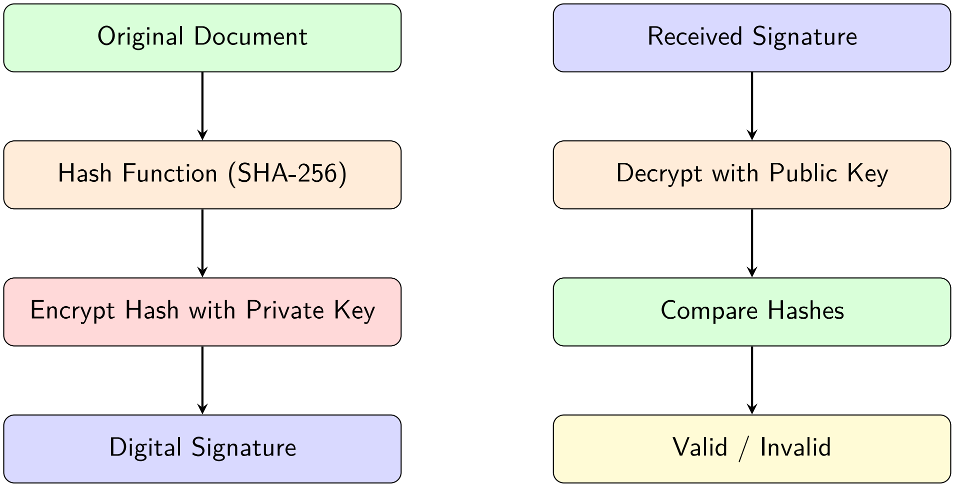
1. **Identify** the signatory
2. Be **uniquely linked** to the signatory
3. Be created using means under the **sole control** of the signatory
4. Detect any **alteration** made to the attached data after signing

Important

Any electronic signature is **ineffective** if the electronic record it's attached to has been **tampered with or amended**.

How Digital Signatures Work

Signing Process | **Verification Process**



Digital Signature vs. Electronic Signature

Feature	**Digital Signature**	**Electronic Signature**
Technology	Public-key cryptography (PKI)	Any electronic mark (typed name, image, click)
Security	High — tamper detection built-in	Varies — depends on implementation
Verification	Mathematically verifiable	May not be verifiable
Legal weight	Stronger under ICT Act 2006	Accepted but weaker
Example	PDF signed with certificate	Typing your name in DocuSign

Public Key Infrastructure (PKI)

What is PKI?

A set of **roles, policies, hardware, software and procedures** needed to create, manage, distribute, use, store and revoke **digital certificates** and manage **public-key encryption**.

Uses of Digital Signatures:

- **SSL/TLS certificates** — Securing web traffic (HTTPS)
- **Code signing** — Verifying software authenticity
- **Document signing** — PDFs, Word documents
- **Email signing** — S/MIME for email authentication
- **Authentication** — Proving identity in systems
- **Git commit signing** — Verifying code contributions

PKI in Bangladesh

Hierarchical Root of Trust Model

Digital signatures issued by a **Certifying Authority (CA)** are considered legally valid.

Structure:

- **CCA (Controller of Certifying Authorities)** under ICT Ministry — the **Root CA**
- **6 Licensed CAs** authorized to issue digital signatures:
  1. Mango Teleservices Ltd
  2. Dohatech New Media
  3. Data-edge Ltd
  4. Banglaphone Ltd.
  5. Computer Services Ltd
  6. Bangladesh Computer Council (BCC)

Role of Certifying Authorities

CA Obligations

- Ensure hardware and software are **safe from intrusion and misuse**
- Provide reasonable **liability** in their service
- Ensure **secrecy and privacy** of digital signatures
- All employees must **comply with rules and regulations**
- **Disclose** information specified in CA rules

Certificate Issuance Requirements

1. Application received from subscriber (via **Registration Authority**)
2. Applicant **properly identified** per certificate practice statement
3. Information to be certified is **accurate**
4. Applicant pays **requisite fees**

Security Requirements for E-Contracts

Confidentiality

Protection of electronic records from **unauthorized disclosure**. Only authorized parties can access contract details.

Authenticity

Parties accessing digital records **are who they claim to be**. Credentials must be recorded and maintained.

Integrity

Electronic records are **not duplicated, modified, or deleted** without authorization.

Non-Repudiation

Parties **cannot deny** having performed an action — e.g., entering a contract, sending a message, or updating a record.

Availability

E-contracting systems and records are **available to authorized parties when required**.

Security Threats to E-Contracts & Digital Signatures

Threats

- **Key compromise** — Theft of private keys
- **Network intrusion** — Man-in-the-middle attacks
- **Software bugs** in PKI deployment
- **Client-side leaks** — Malware on signing device
- **Lack of monitoring** of certificate usage
- **Weak security audits**

Countermeasures

- Regular **security updates** of hardware & software
- Periodic **security review** of the PKI
- Strong **security audits**
- **Certificate Revocation Lists (CRLs)** — Invalidation of compromised certificates
- **Hardware Security Modules (HSMs)** for key storage
- Multi-factor authentication for signing

Key Legislation & Policies

Primary Legislation

ICT Act 2006 — Primary legislation governing digital signatures | e-commerce in Bangladesh

Supporting Rules & Policies

- **Certifying Authority Rules 2010** (CA Rules)
- **National ICT Policy 2018**
- **Certification Practice Statement** (published by the Office of the CCA)
- **Contract Act 1872** — General contract law
- **Evidence Act 1872** — Admissibility of electronic records

Key Takeaways

1. E-contracts are **legally valid** under Bangladesh's ICT Act 2006
2. Digital signatures provide **authenticity, integrity, and non-repudiation**
3. **PKI** is the backbone of digital signature deployment
4. Bangladesh has a **hierarchical CA structure** with CCA as root
5. Not all documents can be signed electronically — know the **exceptions**
6. Security of e-contracts requires addressing **CIA + Non-Repudiation + Availability**
7. As software engineers, you'll build systems that **implement and rely on** these technologies

Discussion / In-Class Activity

Group Discussion (10 minutes)

Form groups of 5–6 students. Discuss one of the following:

1. What challenges might a **Bangladeshi startup** face when implementing e-contracts with international clients?
2. How would you design a **document signing system** that is legally compliant under the ICT Act 2006?
3. What are the risks of using **click-wrap agreements** for SaaS products?

Presentation (2 minutes per group)

Share your findings with the class.